

Physics - Grade 10

Unit 2: Forces and Motion

Lesson 1: Newton's First Law

An object stays at rest or keeps moving at constant velocity unless a net force acts on it.

This property is called inertia.

Lesson 2: Newton's Second Law

The acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass: $F = m a$.

Force is measured in newtons (N), mass in kilograms (kg) and acceleration in m/s^2 .

If the mass increases while the force stays the same, the acceleration decreases.

Example: a net force of 10 N on a 2 kg cart gives an acceleration of 5 m/s^2 .

Lesson 3: Photosynthesis in plants

Green plants make glucose from carbon dioxide and water using light energy.

This process takes place in the chloroplasts.